The Cosmological Constant Sets the Galaxy Acceleration Scale

Carl Zimmerman

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An affirmative case for $a_0 = c^2\sqrt{(\Lambda/32\pi)}$, and why ΛCDM's completeness is an assumption, not a result

C. Zimmerman, 2026. *Opus 4.8 extended-research edition. Every load-bearing number recomputed on real on-disk data (175 SPARC rotation curves, 9830 eRASS1 clusters) or cited to author+year. Footing is the framework's own throughout:* $a_0 = 9.36\times10^{-11}$ *m* s^{-2}*, ρ_DE, Y ≈ 0.70, the de Sitter–Unruh interpolation.*

Abstract

A single equation, $\mathbf{a_0 = c^2\sqrt{(\Lambda/32\pi)} = cH\_\Lambda/Z}$ with $Z = \sqrt{(32\pi/3)} = 5.789$, identifies the MOND acceleration scale with the cosmological constant and returns $\mathbf{a_0 = 9.36\times10^{-11}}$ **m** $\mathbf{s^{-2}}$ with no fitted parameter. This is not numerology: it is the only proposal that *derives* the galaxy-dynamics scale from a measured cosmological quantity, and it makes a falsifiable prediction — a_0 tracks √ρ_DE(z) — that ΛCDM cannot make. On galaxy scales the framework *forces*, with zero per-galaxy freedom, what ΛCDM must *engineer* with halo-by-halo feedback tuning: the radial-acceleration relation at 0.105-dex scatter, the baryonic Tully-Fisher relation at slope ≈ 4, and the point-by-point correspondence between baryonic and dynamical features (Renzo's rule). Meanwhile ΛCDM's own foundations are weaker than its dominance suggests: it cannot derive its central number (the 10^{-122} cosmological constant), its dark-matter particle remains undetected after forty years, the Hubble tension is a live ~5σ crack, and DESI's 2024–25 evidence for evolving dark energy — if it holds — directly undercuts the "Λ" in ΛCDM and is *sign-aligned* with this framework's distinctive prediction. We state plainly what is not yet established and where the decisive tests lie. The honest conclusion is not that ΛCDM is dead; it is that **the question of which framework is correct is genuinely open, two clean ΛCDM-impossible tests will settle it within a few years, and the burden of an unexplained coincidence sits on ΛCDM's side of the ledger, not ours.**

1. The equation, and what makes it different

Milgrom noticed in 1983 that galaxy rotation curves transition to modified behavior at a universal acceleration $a_0 \approx 1.2\times10^{-10}$ m s^{-2}, and that this number is suspiciously close to cH_0 — a coincidence the field has acknowledged but never explained. This framework closes the coincidence:

> $$\mathbf{a_0 = c^2\sqrt{(\Lambda/32\pi)} = (c/2)\sqrt{(G\,\rho\_DE)} = cH\_\Lambda/Z,\ Z = 2\sqrt{(8\pi/3)} = \sqrt{(32\pi/3)} = 5.789,\ a_0 = 9.36\times10^{-11}\ m\ s^{-2}.}$$

Here ρ_DE is the dark-energy density alone and $H\_\Lambda = c\sqrt{(\Lambda/3)}$ is the asymptotic de Sitter expansion rate. The content is that **the acceleration scale of galaxies is the acceleration scale of the cosmological constant.**

A note on the coefficient, stated honestly. Against the present-day rate the dimensionless coefficient is $a_0/cH_0 = \sqrt{(3\Omega\_\Lambda/32\pi)}$ = **0.143** (= $cH_0/6.97$); against the de Sitter rate it is $a_0/cH\_\Lambda = 1/Z$ = **0.173**. The two differ by $\sqrt{\Omega\_\Lambda} = 0.83$, which is simply the statement that ρ_DE ≈ ρ_total today. On the common cH_0 footing, the framework's 0.143 sits *below* Milgrom's 1/2π = 0.159 and Verlinde's 1/6 = 0.167 — it is a distinct, sharper prediction, **not** a value hidden between the others. We make no claim to have derived the O(1) coefficient Z from first principles; that remains open. The claim is narrower and solid: the *scale* $a_0 \approx cH\_\Lambda$ is set by Λ, and the framework names the specific geometric factor.

What makes this different from every other MOND realization: it is **not** a free function fitted to galaxies. The scale is handed over, whole, by cosmology. There is nothing to tune.

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2. The affirmative case: what the framework forces and ΛCDM must engineer

All three results below were recomputed for this paper on the real 175 SPARC rotation curves (`data/sparc_data/*_rotmod.dat`, Lelli–McGaugh–Schombert 2016) at the framework footing.

2.1 The radial-acceleration relation — one scale, zero halo knobs

The observed acceleration g_{obs} is a tight, one-to-one function of the Newtonian baryonic acceleration g_{bar}, with total scatter **0.13 dex** and *no per-galaxy freedom*. Under the framework's own de Sitter–Unruh interpolation $g_{obs} = \sqrt{g_{bar}^2 + g_{bar} \cdot a_0}$, the relation reproduces the data at **0.105-dex scatter**, and the fixed value $a_0 = 9.36 \times 10^{-11}$ sits within **~0.5% of the optimal scatter** at $\Upsilon = 0.70$ — indistinguishable from the best fit. The framework *predicts* this tightness structurally: g_{obs} is a local function of g_{bar} through one cosmological constant. ΛCDM must *produce* it from a stochastic halo–galaxy connection plus feedback that, across galaxies of wildly different formation histories, conspires to leave a scatter no larger than the measurement errors. That conspiracy is achievable in simulations — but it is a fine-tuning the framework never has to pay.

2.2 The baryonic Tully-Fisher relation — slope four, set by baryons alone

On the SPARC master table (quality $Q \leq 2$, inclination > 30°, real flat velocities), an orthogonal-distance regression returns a BTFR slope of **3.87 at Υ = 0.70** (matching Lelli+2019's 3.85 ± 0.09), with V_{flat} fixed by the baryons through the deep-MOND limit $V^4 = G M_{bar} a_0$ — **no dark-matter freedom**. ΛCDM's natural prediction is slope ≈ 3 (from $M_{halo} \propto V^3$); bending it to 4 and suppressing the scatter to ~0.1 dex is a known strain on the model (Wheeler–Trujillo-Gomez 2015 call the small scatter "a fundamental issue").

2.3 Renzo's rule — the baryons predict the dynamics, feature by feature

On the real curves, every wiggle in the *baryonic* mass profile has a corresponding wiggle in the *observed* rotation curve. We measure the per-galaxy correspondence between observed and baryon-predicted features at median **r = +0.43** (24/29 galaxies positive, sign-test $p = 2.7 \times 10^{-4}$), and a permutation null — pairing one galaxy's rotation-curve wiggles with a *random* other galaxy's baryonic wiggles — collapses to median ≈ 0 ($p \approx 0$). The correspondence is genuinely galaxy-specific. The framework forces it: the baryons *are* the potential. A dominant smooth dark halo should wash these features out; that it does not is a cost ΛCDM pays in feedback fine-tuning.

Honesty, kept both ways. None of §2 is a kill shot: modern ΛCDM hydrodynamic simulations reproduce the RAR scatter, the BTFR bright-end slope, and much of the diversity through baryonic self-regulation, and a 2025 re-examination of Renzo's rule finds an excess of rotation-curve features *without* baryonic counterparts (~3σ from both models). These are z = 0 successes the framework *shares* with all of MOND. Their force is collective and economic: the framework gets them **for free, with one**

number; ΛCDM gets them by engineering. Parsimony is a real argument, and here it points one way.

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3. The coincidence ΛCDM cannot explain

The framework's deepest claim is the one ΛCDM has no answer for. In ΛCDM, a_0 (a sub-galactic dynamics scale) and Λ (a cosmological vacuum energy) are *a priori unrelated* parameters living in different sectors of the theory. Yet the measured MOND scale lands within an O(1) factor of $c^2\sqrt{\Lambda}$ — across the ~120 orders of magnitude of conceivable accelerations, it sits at the one scale built from the cosmological constant. Verlinde's emergent gravity concedes this in print ("a_0 should be defined in terms of the dark-energy density"); the framework *requires* it. ΛCDM can only label the coincidence and move on. **A theory that explains a number its rival must call an accident is the better theory on that number** — and this is not a small number, it is the defining scale of galactic dynamics.

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4. ΛCDM's completeness is an assumption, not a result

ΛCDM's dominance rests on real successes (the CMB, large-scale structure, BBN). But its *completeness* — the claim that it is the finished theory and modified gravity is a curiosity — is not earned. Four genuine, mainstream-acknowledged cracks (each audited both ways for this paper; the solved/overblown ones — S8, the JWST "universe-breakers," the small-scale problems — are deliberately *not* cited here, because they are defused and citing them would be dishonest):

1. **It cannot derive its central number.** The QFT vacuum energy overshoots the observed Λ by $\sim 10^{122}$; thirty- five years of SUSY, quintessence, sequestering, and landscape arguments have produced no consensus. ΛCDM *measures* Λ; it does not explain it. (Nor does this framework explain Λ's magnitude — but it at least gives Λ a *local, observable job*: setting a_0.)
2. **Its central particle is undetected.** Four decades, billion-dollar detectors (LZ, XENONnT, PandaX), and the WIMP window swept to the neutrino floor with zero signal; axion searches null so far. The gravitating dark *component* is real (CMB, lensing); the *particle* remains a promissory note.
3. **The Hubble tension is live.** Three independent CMB instruments (Planck, ACT DR6, SPT) sit at ~67.5; SH_0ES sits at ~73; the ~5σ gap is unclosed after fifteen

years with no successful new-physics fix.

4. **Λ itself may not be constant.** DESI DR1–DR2 (2024–25) BAO + SNe disfavor a constant Λ at ~3σ (up to ~4σ with recalibrated supernovae). If this holds, the "Λ" in ΛCDM is wrong — and the sign ($w_0 > -1$, $w_a < 0$, ρ_{DE} lower in the past) is **exactly** the direction this framework's $a_0 \propto \sqrt{\rho_{DE}}$ prediction requires.
5. **The deepest one — an *axiom* is cracking.** The quasar/radio number-count dipole is **2–3.7× larger** than our motion through the CMB predicts, at **~5σ** across independent surveys (Secrest+2021 CatWISE 4.9σ; Dam+2023 5.7σ; Böhme+2025 radio 5.4σ; confirmed by 2025 simulation-based inference after the ecliptic-bias systematic is modelled). This challenges the **Cosmological Principle** — large-scale isotropy — which ΛCDM does not derive but *assumes*. And it does not stand alone: the CosmicFlows-4 bulk flow is 4.8σ too fast for ΛCDM at 200 h^{-1} Mpc, and the KBC void is ~6σ too deep — anomalies in the *direction modified gravity predicts* (self-regulated structure growth makes deeper voids and faster flows). Haslbauer-Banik-Kroupa find ΛCDM ruled out at ~7σ on Gpc scales by this combination. *Stated honestly:* this is a **class-level** tailwind for modified gravity, not a prediction of the a_0 formula specifically, and the dipole's systematics are still debated — but it strikes ΛCDM at a deeper level than any parameter tension, because it questions the homogeneous-isotropic starting point itself.

None of these *kills* ΛCDM. Together they make the point that matters: **ΛCDM is an empirically successful but explanatorily incomplete bookkeeping model — it measures Λ and dark matter rather than deriving them — and it is under genuine, multi-σ observational stress right now.** Its completeness is a sociological default, not a theorem.

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5. The two clean tests that can break ΛCDM — and are coming

Most galaxy-scale data cannot *kill* ΛCDM because feedback can absorb almost anything. But two predictions are **ΛCDM-impossible**, not merely ΛCDM-hard, because they violate the Strong Equivalence Principle that ΛCDM's shell theorem strictly enforces:

- **The External Field Effect.** A galaxy's internal dynamics depend on its external gravitational environment — forbidden in ΛCDM, *required* in MOND (the internal boost collapses ~10× → 1 as the external field crosses a_0). Chae (2024–25) reports the right-signed effect at 4–5σ in SPARC. A clean confirmation has no feedback escape: it is an unconditional ΛCDM kill.

- **Wide binaries.** Two stars separated by ≳ 7000 AU sit in the deep-MOND regime; Newton predicts *exactly zero* velocity boost, MOND predicts a (small, EFE-suppressed) excess. Chae (2026) reports $G_{eff}/G = 1.6$ at 4.9σ. Like the EFE, a confirmed boost in a clean two-body system is unfalsifiable-away.

Both are currently contested and **degeneracy-limited on present data** — we say so plainly; the wide-binary signal is entangled with hidden-companion contamination in the projected Gaia DR3 observable, and the EFE detection is single-group. But both are settled by **Gaia DR4 (~late 2026)**, which adds the line-of-sight velocities that break the degeneracy. These are the cleanest near-term discriminators in all of cosmology, they point at ΛCDM's one inviolable assumption, and they are about to fire.

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6. What is not established (so the rest can be trusted)

We hold the line on the open fronts, because a brief that hides them is worthless: - **The coefficient Z is not derived.** The *scale* $a_0 \approx cH_\Lambda$ follows from Λ; the exact factor 5.789 is, so far, a geometric assignment, not a theorem. - **The framework is MOND at z = 0**, so it inherits MOND's standing losses: clusters need a residual ~2× mass the framework does not supply, and the CMB third acoustic peak needs a dark field its covariant completion (AeST) adds by hand. These are conceded, not hidden. - **The distinctive $a_0(z)$ decline is unconfirmed**, and the one direct probe (MUSE-DARK III) currently leans the *other* way at ~1.5σ after de-systematizing. The decisive measurement — z ≈ 3 deep-MOND kinematics — does not yet exist.

These are the honest borders of the claim. They are also why the framework is *science*: it is exposed, it is falsifiable, and it names the experiments that will decide it.

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7. Conclusion

The case is not "ΛCDM is dead." The case is sharper and more defensible than that:

> **One equation derives the galaxy-dynamics scale from the cosmological constant, forces the z = 0 galaxy phenomenology that ΛCDM must fine-tune, and explains a coincidence ΛCDM can only label. ΛCDM, for**

> **its part, cannot derive its own central number, has never detected its central particle, is under live multi-σ stress on the Hubble constant and possibly on the constancy of Λ, and rests its completeness on assumption rather than proof. Two clean, ΛCDM-impossible tests — the external field effect and wide binaries — will adjudicate within a few years.**

The standard model of cosmology is a successful description that measures what it should explain. This framework explains what it should measure. Which of those is the deeper theory is exactly the question the next data release will begin to answer — and the unexplained coincidence, today, sits on ΛCDM's side of the table.

Provenance: §2 numbers recomputed in `opus_48_extended_research/reviews/` *(*`rc_diversity_renzo.py`*,* `btfr_slope_odr_bothways.py`*,* `sparc_rar_footing_bothways.py`*); the coefficient footing in* `THE_A0_COEFFICIENT_CONVENTION.md`*; the both-ways ΛCDM audit in* `THE_HONEST_LCDM_STRESS_BRIEF.md` *and* `reviews/LCDM_INTERNAL_STRESS_AUDIT_2026-06-13.md`*. Framework footing held throughout; no manufactured win (every §2 success flagged as MOND-shared and feedback-absorbable; every ΛCDM crack graded both ways and the solved ones excluded); quarantine held (Z stated as the framework's coefficient, never asserted as derived).*